

Mock Exam Solutions

SECTION A: ATTEMPT ALL QUESTIONS (55 marks)

1. Match Column A to Column B (5 marks)

Column A	Column B
A. The conics has a Centre, the semi major axis is always greater than the semi minor axis.	V. Ellipse
B. It has an eccentricity greater than 1.	iv. Hyperbola
C. It is a set of equidistant points from a fixed point and a fixed line.	i. The parabola
D. It displays the Centre, the semi major axis and the semi minor axis of conics.	iii. Canonical/standard equation form of conics
E. It has a focus at pole and the directrix at the focal parameter from the pole.	ii. Polar equation form of conics

2. Probability of Committee Selection (3 marks)

Given: 6 men, 8 women $\rightarrow$ total 14, Choose 5 people

i. Exactly 3 men

$$\text{Ways} = \binom{6}{3} \times \binom{8}{2} = 20 \times 28 = 560, \quad \text{Total} = \binom{14}{5} = 2002, \quad P = \frac{560}{2002} = \frac{40}{143}$$

Answer: b. $\frac{40}{143}$

ii. At least 2 women

$$\text{Cases: } (2W, 3M) + (3W, 2M) + (4W, 1M) + (5W, 0M) = 560 + 840 + 420 + 56 = 1876, \quad P = \frac{1876}{2002} = \frac{134}{143}$$

Answer: a. $\frac{134}{143}$

iii. More women than men

Cases: $(3W, 2M) + (4W, 1M) + (5W, 0M) = 840 + 420 + 56 = 1316$, $P = \frac{1316}{2002} = \frac{94}{143}$

Answer: c. $\frac{94}{143}$

3. Integral Evaluation (4 marks)

Evaluate:

$$\int_1^{e^2} \ln x \, dx$$

By parts:

$$= x \ln x - x \Big|_1^{e^2} = e^2(2) - e^2 - (0 - 1) = e^2 + 1$$

Answer: None of the options match. Correct value is $\boxed{e^2 + 1}$

4. Conic with Two Foci (6 marks)

Given: Foci at $(-5, 2)$ and $(1, 2)$, $e = \frac{2}{3}$

i. Type of Conic: $e < 1 \Rightarrow$ Ellipse

Answer: d. Ellipse

ii. Centre, vertices, covertices:

- Centre: $(-2, 2)$
- $c = 3, a = \frac{9}{2}, b = \frac{3\sqrt{5}}{2}$
- Vertices: $(-6.5, 2), (2.5, 2)$
- Covertices: $(-2, 2 \pm \frac{3\sqrt{5}}{2})$

iii. Graph Description:

Plot center, foci, vertices, covertices. Directrices at $x = -8.75$ and $x = 4.75$

5. Linear Combination of Vectors (2 marks)

Find y so that $(1, y, 5)$ is in span of $(1, -3, 2), (2, -1, 1)$

Solution:

$$c_1 = 3, c_2 = -1 \Rightarrow y = -3(3) - (-1) = -8$$

Answer: c. $y = -8$

6. Complex Number Equation (4 marks)

Solve:

$$z^2 + (5 - i)z + 8 - i = 0$$

Using quadratic formula:

- Discriminant: $-8 - 6i$
- $\sqrt{-8 - 6i} = \pm(1 - 3i)$

Solutions:

$$z_1 = -2 - i, \quad z_2 = -3 + 2i$$

Answer: c. $\{-2 - i, -3 + 2i\}$

7. Tangent Line to Ellipse (4 marks)

Given:

$$x + 2y = k \text{ tangent to } x^2 + 4y^2 = 8$$

Tangency condition $\Delta = 0 \Rightarrow k = \pm 4$

Answer: a. ± 4

8. Loudness Increase (3 marks)

If $I \rightarrow 100I$,

$$L = 10 \log \left(\frac{100I}{I_0} \right) = 20 + L_1$$

Answer: b. 20 dB

9. Binomial Probability (2 marks)

$X \sim B(10, 0.5)$, $P(X = 5) = \frac{63}{256} \approx 0.246$

Answer: a. 0.246

10. Discrete Random Variable (6 marks)

X	6	7	8	9
P(X)	0.1	0.2	0.3	0.4

- $P(X = 8) = 0.3$ (b)
- $P(X > 7) = 0.7$ (a)

- $E(X) = 8$ (c)
- $Var(X) = 1$ (c)
- $E(5 - 2X) = -11$ (b)
- $Var(5 - 2X) = 4$ (c)

11. Arc Length of Curve (3 marks)

Given $y = \cosh x, y' = \sinh x$

$$\text{Length} = \int_0^2 \cosh x \, dx = \sinh 2 = \frac{e^2 - e^{-2}}{2}$$

Answer: b. $\frac{e^2}{2} - \frac{1}{2e^2}$

12. Differential Equation (4 marks)

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} + \frac{1}{4}y = 0 \Rightarrow r = \frac{1}{2}, \quad y = (C_1 + C_2x)e^{\frac{1}{2}x}$$

Answer: c. $(C_1 + C_2x)e^{\frac{1}{2}x}$

13. Integral with Absolute Value (3 marks)

$$\int_0^{1/2} (1 - 2x)dx + \int_{1/2}^2 (2x - 1)dx = \frac{1}{4} + \frac{9}{4} = \frac{5}{2}$$

Answer: b. $\frac{5}{2}$

14. Limit and Parameter (2 marks)

$$\lim_{x \rightarrow -\infty} \left(\frac{a + \frac{1}{x}}{a - \frac{1}{x}} \right)^{-x} = e^{-1} \Rightarrow e^{\frac{-2}{a}} = e^{-1} \Rightarrow a = 2$$

Answer: b. 2

15. Dissolving Sugar (4 marks)

$$M(t) = 50e^{-kt}, \quad 20 = 50e^{-5k} \Rightarrow k \approx 0.1832$$

$$\text{To reach 5 kg: } t = \frac{\ln(10)}{0.1832} \approx 12.564, \quad \text{Extra time: } 12.564 - 5 \approx 7.564$$

Answer: ~ 7.564 hours

SECTION B: ATTEMPT ONLY THREE QUESTIONS (45 marks)

16. Committee and Book Selection (15 marks)

a)

i. $\binom{9}{3} = 84$

ii. $\binom{4}{1} \cdot \binom{5}{2} = 40$

iii. $\binom{4}{2} \cdot \binom{5}{1} = 30$

b)

i. $\frac{1 \cdot \binom{8}{2}}{\binom{9}{3}} = \frac{28}{84} = \frac{1}{3}$

ii. $\binom{5}{2} \cdot \binom{3}{1} = 30 \Rightarrow \frac{5}{14}$

17. Complex Locus and Integrals (15 marks)

a)

$$\left| \frac{z-1}{z+1} \right| = 2 \Rightarrow \text{Circle centered at } \left(-\frac{5}{3}, 0 \right), r = \frac{4}{3}$$

b)

• $I_0 = \frac{\pi}{2}, I_1 = 1, I_3 = \frac{2}{3}$

• Recurrence: $I_n = \frac{n-1}{n} I_{n-2}$

• $I_4 = \frac{3\pi}{16}, I_5 = \frac{8}{15}$

18. Parametric Curve and Integral (15 marks)

a)

Area enclosed:

$$36 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = 108\pi$$

b)

Use identities to get:

$$\int \sin^2 x \cos 3x dx = \frac{\sin 3x}{6} - \frac{\sin 5x}{20} - \frac{\sin x}{4} + C$$

19. Hyperbola Analysis (15 marks)

Given: $16x^2 - 9y^2 - 96x + 279 = 0$

- Standard form: $\frac{y^2}{15} - \frac{(x-3)^2}{135/16} = 1$
- Center: $(3, 0)$, Vertices: $(3, \pm\sqrt{15})$
- Foci: $(3, \pm\frac{5\sqrt{15}}{4})$
- Asymptotes: $y = \pm\frac{4}{3}(x - 3)$

20. Statistical Analysis (15 marks)

X	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$	R_x	R_y	$d = R_x - R_y$	d^2
9	10	0	-2.6	0	6.76	0	5.5	4	1.5	2.25
12	13	3	0.4	9	0.16	1.2	7.5	5.5	2	4
5	8	-4	-4.6	16	21.16	18.4	2	2.5	-0.5	0.25
6	20	-3	7.4	9	54.76	-22.2	3.5	10	-6.5	42.25
9	13	0	0.4	0	0.16	0	5.5	5.5	0	0
14	17	5	4.4	25	19.36	22	9.5	9	0.5	0.25
3	5	-6	-7.6	36	57.76	45.6	1	1	0	0
6	8	-3	-4.6	9	21.16	13.8	3.5	2.5	1	1
12	16	3	3.4	9	11.56	10.2	7.5	7.5	0	0
14	16	5	3.4	25	11.56	17	9.5	7.5	2	4
Sum				138	205.4	96				54

a) Table includes:

- $\sum(x - \bar{x})^2 = 138$
- $\sum(y - \bar{y})^2 = 205.4$
- $\sum(x - \bar{x})(y - \bar{y}) = 96$
- $\sum d^2 = 54$

b)

- Variance X = 13.8
- Variance Y = 20.54
- Covariance = 9.6
- Pearson $r = 0.570$

c) Spearman $\rho = 0.6727$

d) Regression line:

$$\hat{y} = 6.339 + 0.696x$$